

UNIVERSITI KUALA LUMPUR

YEAR: 2025

SEMESTER: OCTOBER

COURSE CODE:

ISB 26603 / IIB26603


UniKL
 UNIVERSITI
 KUALA LUMPUR

UniKL MIIT
COURSE LEARNING PLAN
To be shared with students.

SECTION A: COURSE DETAILS

1.	Name of Course : MOBILE AND UBIQUITOUS COMPUTING Course Code : ISB 26603 / IIB26603																												
2.	Synopsis : This course aimed at providing students to understand the state of the art of mobile and ubiquitous computing technologies, and to get familiar with the development of application software for mobile application. Mobile and ubiquitous computing will be discussed from three perspectives: mobile technology, application development, and user interaction. The course will start overview various mobile and ubiquitous computing applications, technologies and wireless communication. Next, students will learn about common paradigms in ubiquitous computing such as smart devices, smart environment, and smart interactions. Students will be introduced to and use mobile application frameworks and development environments to reinforce concepts covered in lectures. User interface and user experience will be discussed and application development guidelines from various vendors will be discussed and analyzed. Lastly, the course will look at some current research in mobile computing. Students will be expected to learn at least one mobile application development framework and use it to implement their assignments and course project.																												
3.	Name(s) of academic staff : TS. NOR AZLINA ALI, TS. NURDATILLAH HASIM, DR. CHEN XINYUAN																												
4.	Semester and Year offered : <table border="1" style="width:100%; text-align:center;"> <tr> <td>Semester</td> <td>2</td> <td>Year</td> <td>2</td> </tr> </table>													Semester	2	Year	2												
Semester	2	Year	2																										
5.	Credit Value : 3																												
6.	<table border="1" style="width:100%; text-align:center;"> <tr> <th>Student Learning Time (SLT):</th> <th>Lecture (F2F)</th> <th>Tutorial (F2F)</th> <th>Practical (F2F)</th> <th>Others (F2F)</th> <th>Guided / Online / Technology-mediated (NF2F / Synchronous)</th> <th>Independent Learning (N-F2F)</th> <th>TOTAL</th> </tr> <tr> <td></td> <td>17</td> <td>-</td> <td>34</td> <td>-</td> <td>9</td> <td>60</td> <td>120</td> </tr> </table>													Student Learning Time (SLT):	Lecture (F2F)	Tutorial (F2F)	Practical (F2F)	Others (F2F)	Guided / Online / Technology-mediated (NF2F / Synchronous)	Independent Learning (N-F2F)	TOTAL		17	-	34	-	9	60	120
Student Learning Time (SLT):	Lecture (F2F)	Tutorial (F2F)	Practical (F2F)	Others (F2F)	Guided / Online / Technology-mediated (NF2F / Synchronous)	Independent Learning (N-F2F)	TOTAL																						
	17	-	34	-	9	60	120																						
7.	Prerequisite/co-requisite: (if any) ISB16003 Object Oriented Programming																												
8.	Course Learning Outcomes (CLO): At the end of the course the students will be able to... (example) explain the basic principles of immunisation (C2,PLO1)																												
	CLO1 Identify and solve problems based on the leading theory of mobile and ubiquitous computing (C2, PLO7)																												
	CLO2 Analyze how smart devices, smart environment, and smart interaction works in mobile and ubiquitous computing (C4, PLO9)																												
	CLO3 Design and develop a mobile application (P3, PLO8)																												
9.	Mapping of the Course Learning Outcomes to the Programme Learning Outcomes, Teaching Methods and Assessment : Please select the Learning Outcome Domain (LOD) for each PLO in the cells above it. E.g. PLO1 - Knowledge, PLO2 - Cognitive, PLO3 - Practical Skills																												
	Programme Learning Outcomes (PLO) - BSE / BCSS												Learning and Teaching Method	Assessment Method															
	Digital skills	Personal skills	Communication skills	Cognitive skills	Ethics and professionalism	Entrepreneurial skills	Knowledge and understanding	Practical skills	Numeracy skills																				
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12																	
	CLO 1	√												Lecture, Lab	Quiz, Test														
	CLO 2			√										Lecture, Lab	Assignment														
	CLO 3		√											Lecture, Lab	Mini Project, Final														
	Programme Learning Outcomes (PLO) - BNS												Learning and Teaching Method	Assessment Method															
	Knowledge and understanding	Practical skills	Interpersonal skills	Ethics and professionalism	Communication skills	Cognitive skills	Digital skills	Entrepreneurial skills	Numeracy skills																				
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12																	
	CLO 1						√							Lecture, Lab	Quiz, Test														
	CLO 2								√					Lecture, Lab	Assignment														
	CLO 3							√						Lecture, Lab	Mini Project, Final														
	Programme Learning Outcomes (PLO) - BIOT												Learning and Teaching Method	Assessment Method															
	Knowledge and understanding	Practical skills	Numeracy skills	Cognitive skills	Practical skills	Personal skills	Personal skills	Ethics and professionalism	Leadership, autonomy and responsibility	Communication skills	Entrepreneurial skills	Digital skills																	
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12																	
	CLO 1	√												Lecture, Lab	Quiz, Test														
	CLO 2			√										Lecture, Lab	Assignment														
	CLO 3		√											Lecture, Lab	Mini Project, Final														

SECTION A: COURSE DETAILS

Course Learning Outcomes (CLO)	Programme Learning Outcomes (PLO) - NIL												Learning and Teaching Method	Assessment Method
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12		
CLO 1														
CLO 2														
CLO 3														
10	Transferable Skills (if applicable) (Skills learned in the course of study which can be useful and utilized in other settings)						1	Digital skills						
							2	Interpersonal skills						
							3							
							4							
							5							
11	Assessments	Type				Methods				Weightage (%)				
		Continuous Assessment(s)				Quiz				10				
						Assignment				10				
						Test				20				
						Mini Project				20				
						Final Test				40				
		Final Assessment(s)												
Total				5 assessment methods				100%						
12	References (include required and further readings that are most current)						Required reading:	Lawrence T. Yang, Evi Syukur, Seng W. Loke. Handbook on Mobile and Ubiquitous Computing Taylor & Francis, CRC Press. 2016. [eBook ISBN 9781439848128]						
							Further readings:	2. Anubhav Pradhan, Anil V. Deshpande, Composing Mobile Apps, Learn, Explore Apply using Android, 1st Edition, Wiley 2014. [ISBN: 978-81-265-4660-2] 3. Erik Hellman, Android Programming: Pushing the Limits, Wiley, 2014. [ISBN 978-1-118-71737-0] 4. A. Genco and S. Sorce, Pervasive Systems and Ubiquitous Computing, 1st Edition, Hardcover – May 21, 2010.						
13	Other additional information :						Course effective date: 02/01/2019							


SECTION B: LEARNING SCHEDULE

Week		Topics	CLO	Delivery Methods	Assessments	
WEEK 1	9/10 - 10/10	Ubiquitous Computing Concepts, Architecture and Applications	1	Lecture		
WEEK 2	13/10 - 17/10	Mobile Application Environment Lab 1: Setting mobile application development environment	1	Lecture and lab	Lab Exercise	Lab Setup (default app counter) https://docs.flutter.dev/get-started/install
WEEK 3	20/10 - 24/10	Mobile Application Design - Human Computer Interaction Lab 2: Introduction to mobile application development	1	Lecture and lab	Lab Exercise	
WEEK 4	27/10 - 31/10	Mobile Application - Graphics and Multimedia Lab 3: Mobile Application- UI and UX	1,2	Lecture and lab	Assignment**	https://docs.flutter.dev/ui/widgets
WEEK 5	3/11 - 7/11	Mobile Application - Context Aware and Smart Environment Lab 3: Mobile Application- UI and UX	1,2,3	Lecture and lab	Lab Exercise Quiz **	MDC102 + QUIZ
WEEK 6	10/11 - 14/11	Mobile Application - Data Management Lab 4: Mobile Application - Multimedia, graphic and Animation	1,2,3	Lecture and lab	Assignment Demonstration & Presentation	MDC103
WEEK 7	17/11 - 21/11	Lab 4: Mobile Application - Multimedia, graphic and Animation	1,2,3	Lecture and lab	Lab Exercise Test **	MDC 104 + TEST
WEEK 8	1/12 - 5/12	Lab 5: Mobile Application - Application Program Interface (API)	1,2,3	Lecture and lab	Lab Exercise	Firebase set up
WEEK 9	08/12 - 12/12	Lab 5: Mobile Application - Application Program Interface (API)	1,2,3	Lecture and lab	Lab Exercise	
WEEK 10	15/12 - 19/12	Lab 6 : Mobile Application - Database	1,2,3	Lecture and lab	Mini Project	Project briefing
WEEK 11	22/12 - 26/12	Lab 6 : Mobile Application - Database	1,2,3	Lecture and lab		Project progress monitoring
WEEK 12	29/12 - 2/1	Mini project Preparation	1,2,3	Presentation	Mini Project Demonstration & Presentation	PROJECT DEMO
WEEK 13	5/1 - 9/1	Mini project Preparation	1,2,3	Presentation	Mini Project Submission	PROJECT DEMO
WEEK 14	12/1 - 16/1				FINAL TEST	FINAL
		Note: ** Subject to change				